# Experiment Plan: Model-Native Token Selection for Video-LLMs

Phronetic AI — ML Team · Companion plan for *"What does the VLM See?"*

## Guiding principle

Run the experiments that can kill the idea cheaply *before* building the full pipeline. The classic failure mode is to spend weeks training the scorer and wiring up the benchmark sweep, then discover the supervision signal was never strong enough. The Phase 0 experiments below are training-free (forward passes only). Together they either green-light the project or tell you exactly what to fix: which layers, whether the scorer must be query-conditioned, whether the attention signal tracks causal importance, and whether the Pareto premise behind budgeting holds.

> **Do not start by training the scorer.** Start with B and C. They cost almost nothing and determine whether the paper's core claims are even available to you.

## Shared setup

Lock these down once so every experiment is directly comparable.

| Item | Fix before starting |
|---|---|
| Backbone | Qwen2.5-VL-7B-Instruct, frozen, throughout. |
| Dev set | ~300–500 NExT-QA val QA pairs (primary) + a 200-pair VideoMME slice (cross-check). Small enough to iterate in hours. |
| Retention ratios | $\rho \in \{10\%, 20\%, 25\%, 50\%\}$. Include the aggressive end (10–20%); 75% carries little signal. |
| Frame / patch budget | Fix $T$ (frames) and $N$ (patches/frame) now and report them, so token counts and speedups are comparable across experiments. |
| Primary metric | Multiple-choice top-1 accuracy. Add wallclock later, including scorer + sort + budgeting overhead. |

## Phase 0 — Validation (training-free, run first)

### Experiment A — Critical-layer localization on your own model

**Goal:** Replace the imported "layers 12–16" with a measurement on Qwen2.5-VL-7B itself. The source finding (Gou et al., CVPR 2026) localizes critical layers with a sliding-window knockout and reports them as model-dependent, not a universal range.
**Procedure:** Apply a sliding-window language-to-video attention knockout (window of 2–4 layers) across all 28 decoder layers. For each window, measure the multiple-choice accuracy drop on the dev set.
**Output:** An accuracy-drop-vs-depth curve. Set the method's layer set $\mathcal{L}$ to the window(s) with the largest drop.
**Decision:** If the peak is not near 12–16, your fixed layer set changes. Far better to learn this now than to defend an imported number in review.
**Cost:** 1–2 days. No training.

### Experiment B — The oracle ceiling (most important early test)

**Goal:** Establish the upper bound your distilled scorer can ever reach, and confirm the supervision signal beats trivial baselines.
**Procedure:** Use the teacher scores directly — full language-to-video attention from the layers chosen in A — to select top-$k$ tokens at inference. Measure accuracy at each $\rho$. Compare against uniform sampling and KiToke at matched retention. (Not deployable; it needs the full forward pass. That is fine — it is a ceiling.)
**Output:** Accuracy vs. $\rho$ for attention-oracle, uniform, and KiToke.

> **Kill criterion:** If the attention-oracle barely beats uniform sampling, the signal carries no usable information and no amount of distillation will save it — **stop and pivot**. If it clearly beats uniform and KiToke, you have a real bound to chase and everything downstream is justified.

**Cost:** Forward passes only; a few days.

### Experiment C — How query-dependent is the optimal selection?

**Goal:** Settle the central design question from the review: must the scorer see the question, or can it stay query-blind? NExT-QA has multiple questions per video, which makes this directly measurable.
**Procedure:** For each video, compute the teacher top-$k$ token set per question and measure Jaccard overlap across questions. Then build a query-blind oracle (average teacher scores over all of a video's questions; pick one fixed top-$k$ set) and compare its per-question accuracy against the per-question oracle from B.
**Output:** Cross-question overlap distribution, and the accuracy gap between query-blind and per-question oracles. That gap is the value of query conditioning.
**Decision:** High overlap / small gap $\Rightarrow$ a query-blind MLP on patch embeddings is fine; reframe honestly as a distilled saliency predictor. Low overlap / large gap $\Rightarrow$ the scorer must be conditioned on the question (e.g. cross-attention to query embeddings), because a query-blind student structurally cannot capture this.
**Cost:** Near-zero; reuses B's forward passes.

### Experiment D — Does attention magnitude track causal importance?

**Goal:** Validate the premise that the quantity you distill (attention weight) is the quantity you care about (effect on the answer).
**Procedure:** On a small set, compare each token's teacher attention score to a leave-one-out causal effect: the drop in the gold-answer logit when that token (or a small spatial group) is removed. Use grouped ablations to stay tractable. Report Spearman correlation.
**Output:** Correlation between attention-derived importance and causal importance.
**Decision:** Weak correlation means you are distilling a correlational proxy that doesn't move accuracy — switch the supervision target (e.g. to a knockout-derived signal) or rethink the layers.
**Cost:** Moderate; keep the sample small.

### Experiment E — Pareto premise & estimator choice (added: validates the budgeting contribution)

> Section 2.4 *asserts* that token importance is Pareto and derives per-bin budgets from its shape. Three cheap checks (forward passes only, then numpy on the extracted scores) decide whether that contribution stands — run them before writing the MLE derivation as the centrepiece.

**Goal:** Confirm (i) the importance *tail* is genuinely Pareto rather than lognormal all the way up,

(ii) the shape varies enough across temporal bins for adaptive budgeting to beat uniform, and (iii) which estimator to trust at realistic bin sizes.
**Procedure:**

1. **Tail fit.** Pool teacher scores $s_i$ per bin; run the Clauset–Shalizi–Newman procedure — fit $x_{\min}$ by KS minimization, bootstrap a $p$-value, and likelihood-ratio test power-law vs. *lognormal* vs. exponential. The lognormal comparison is the one that matters, since the paper says the bulk is lognormal.

2. **Shape variation.** Plot the distribution of per-bin $\widehat{\alpha}_t$ (Hill) and $G_t$ (Gini). If they are near-constant, adaptive budgeting collapses to uniform and there is no headroom — this single plot tells you whether the contribution exists.

3. **Hill vs. Gini stability.** Subsample bins to realistic per-bin token counts; compare the variance/bias of $\widehat{\alpha}$ (Hill) against $\alpha = (G{+}1)/2G$ (Gini) versus a large-sample reference. The $\mathrm{Var}(\widehat{H}) = 1/(\alpha^2 k)$ result predicts Gini should be tighter at small $k$.

**Output:** Tail-fit $p$-values and LR statistics; per-bin shape histograms; a Hill-vs-Gini variance curve.
**Decision:** If the tail is not distinguishably Pareto, reframe "Pareto-adaptive" as "concentration-adaptive" and lead with Gini. If shapes barely vary, drop adaptive budgeting in favour of stratified-uniform. If Gini is materially tighter at small $k$, adopt it as the deployed estimator and cite Hill as the asymptotic ideal.
**Cost:** Reuses B's forward passes + a day of analysis. No training.

## Phase 1 — Component validation (after Phase 0 passes)

### Norm-asymmetry experiment (contribution #4)

Cheap, standalone, and it motivates the learned scorer in the paper's own narrative. Select tokens by top-$k$ vs. bottom-$k$ of the L2 norm of ViT features at each $\rho$, and show that bottom-$k$ norm selection is stable (reliably discards uninformative tokens) while top-$k$ norm selection degrades accuracy (fails to find important ones). Run it early.

> **Note:** this also stress-tests Pareto budgeting if budgets are derived from *norms*. If norms can't rank importance, the budgeting must instead be computed on the importance scores (scorer logits / teacher attention), or that component needs separate justification.

### Distillation fidelity

Before spending compute on downstream evaluation, train the scorer (the query-blind variant and, if Experiment C demands it, the query-conditioned variant) and measure how well it reproduces the teacher ranking — top-$k$ recall and NDCG against the teacher scores. This isolates "does the MLP learn the signal" from "does the signal help," so a later downstream miss is diagnosable.

## Phase 2 — Main results and ablations (only after Phases 0–1 succeed)

**Main table:** Full / Uniform / L1-delta / FastV / DyCoke / KiToke / LearnPruner / SCORE / Ours, across $\rho$ and {VideoMME, MVBench, EgoSchema}. Report accuracy *and* wallclock — and include scorer + sort + budgeting overhead in the wallclock, not just LLM FLOPs. FastV and LearnPruner are non-optional: FastV is your stated foil, and LearnPruner is your closest methodological neighbour.

**Ablations:** Supervision source (language-to-video vs. CLS vs. all-token-average vs. random); layer range (informed by A); and a three-way selection comparison — global top-$k$ vs. uniform-per-bin vs. Pareto-adaptive — paired with a temporal-coverage metric (e.g. entropy of retained tokens over bins) so the budgeting component is validated rather than assumed.

### Decision gates at a glance

| Exp. | Question it answers | Decision / kill criterion |
|---|---|---|
| A | Which layers are actually critical in Qwen2.5-VL-7B? | If the peak is far from 12–16, the fixed layer set changes. Resolve now, not in rebuttal. |
| B | What is the ceiling? Does the teacher signal beat simple baselines? | If the attention-oracle barely beats uniform sampling, the supervision carries no usable signal — **STOP and pivot**. |
| C | How query-dependent is the optimal token set? | Large per-question gap ⇒ scorer *must* be query-conditioned. Small gap ⇒ query-blind MLP is fine; reframe as a saliency predictor. |
| D | Does attention magnitude track causal importance? | Weak correlation ⇒ you are distilling a proxy that doesn't move accuracy; change the supervision target. |
| E | Is the tail Pareto, does shape vary, which estimator? | Not Pareto ⇒ reframe as concentration-adaptive. Flat shapes ⇒ drop adaptive budgeting. Gini tighter ⇒ deploy Gini. |

### Sequencing

1. A and B first, in parallel if resources allow — they define the layers and the ceiling.
2. C immediately after B (reuses its forward passes) — decides query-conditioning before any architecture is committed.
3. D alongside B/C — confirms the attention→causal premise.
4. E reuses B's scores — confirms the Pareto/budgeting premise.
5. **Gate:** proceed only if B clears its kill criterion.
6. Norm-asymmetry + distillation fidelity (Phase 1).
7. Main table + ablations (Phase 2).

**The single most important point:** B and C are cheap and decide whether the paper's core claims are reachable. Run them before committing to the scorer.